Mid-term Exam II

Review of Modules 4–6

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

Mid-term Exam II

- Monday 10/26, in class, 50 minutes, 100 points, covers **Modules 4–6** only
- Questions 1–10 multiple choice (6 points each), Questions 11–12 free response (20 points each)
- Show your work on free response, partial credit for correct steps
- Formula sheet provided, plus your **financial calculator** and one self-prepared cheat sheet (one page, double-sided, A4)

Set the Number of Digits to at Least 4!

- On the TI BA II Plus, press
  - **2nd**
  - .
  - 4
  - **ENTER**

Set the Number of Compounding Periods per Year to 1!

- On the TI BA II Plus, press
  - **2nd**
  - **I/Y** (**P/Y**)
  - 1
  - **ENTER**

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  - **ENTER**

You must do this!

Financial Calculators, Use Case One



- Financial calculators solve any one of **N**, **I/Y**, **PV**, **PMT**, and **FV** from the other four

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- Your financial calculator is programmed to solve this equation

$$\text{PV} + \left[\frac{\text{PMT}}{(1+I)^1} + \frac{\text{PMT}}{(1+I)^2} \cdots + \frac{\text{PMT}}{(1+I)^N}\right] + \frac{\text{FV}}{(1+I)^N} = 0$$

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- **IRR**, your financial calculator solves this equation for I

$$\mathrm{CF}_0 + \frac{\mathrm{CF}_1}{(1+I)^1} + \frac{\mathrm{CF}_2}{(1+I)^2} + \cdots + \frac{\mathrm{CF}_N}{(1+I)^N} = 0$$

Use Case Two, Keys for NPV and IRR

NPV

Enter	Keys
	CF, **2nd**, **CLR Work**
CF_0	**ENTER**, ↓
CF_1	**ENTER**, ↓, ↓
⋮	⋮
CF_N	**ENTER**
	NPV (the calculator asks for I)
I	**ENTER**, ↓
	CPT
NPV	

IRR

Enter	Keys
	CF, **2nd**, **CLR Work**
CF_0	**ENTER**, ↓
CF_1	**ENTER**, ↓, ↓
⋮	⋮
CF_N	**ENTER**
	IRR (shows 0)
	CPT
IRR	

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- -1 **PV**, 6 **I/Y**, 0 **PMT**, 2 **FV**, then **CPT** **N**

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$$N = ?, \quad I = 6\%$$

- -1 **PV**, 6 **I/Y**, 0 **PMT**, 2 **FV**, then **CPT** **N**
- The output is 11.9, about 11.9 years, because $(1 + 0.06)^{11.9} \approx 2$

Recap, Three Kinds of Returns

Return	Holding period return	Annualized return
Percentage growth of an asset's value over some period of time	Percentage growth over the whole period an asset is owned, longer or shorter than a year	Percentage growth per year, a geometric mean over the T years of the holding period

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- E.g., a stock returns 80% in the first year and -20% in the second year
  - Holding period return, $1.80 \times 0.80 - 1 = 44\%$

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  - Holding period return, $1.80 \times 0.80 - 1 = 44\%$
  - Annualized return, $1.44^{1/2} - 1 = 20\%$, *not* $(80\% - 20\%)/2 = 30\%$

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- On the calculator, N **N**, r **I/Y**, C **PMT**, 0 **FV**, then **CPT** **PV**
- Timing, the first payment is at $t = 1$. Due (first payment at $t = 0$), multiply by $(1 + r)$. Delayed (first payment at $t > 1$), multiply by $1/(1 + r)^{t-1}$

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$$(1 + \text{Real return}) = \frac{(1 + \text{Nominal return})}{(1 + \text{Inflation rate})}$$

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- E.g., a nominal return of 6% and an inflation rate of 2%, real return $= 1.06/1.02 - 1 = 3.9\%$, approximately $6\% - 2\% = 4\%$

Practice Problem

- You have $20,000 in the bank earning interest of 5% per year. You need $30,000 to make a down payment on a house. You can save an additional $1,000 per year, deposited at the end of each year. How long will it take you to accumulate the $30,000?

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$I = 5\%, \quad N = ?$

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- Solution, we need to solve for **N**,
  - $-20{,}000$ **PV** (the money already in the bank)
  - $-1{,}000$ **PMT** (the yearly deposits)
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- The calculator is solving

$$-\$20{,}000 - \left[\frac{\$1{,}000}{(1+0.05)^1} + \cdots + \frac{\$1{,}000}{(1+0.05)^N}\right] + \frac{\$30{,}000}{(1+0.05)^N} = 0$$

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- The output is 4.57, so it takes about 4.6 years to accumulate the \$30,000

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- **Bond value**, the present value of the coupons and the face value, discounted at the required rate of return r

$$\text{Value} = \frac{\text{Coupon}}{(1+r)^1} + \frac{\text{Coupon}}{(1+r)^2} + \cdots + \frac{\text{Coupon} + \text{Face value}}{(1+r)^T}$$

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- On the calculator, T **N**, r **I/Y**, coupon **PMT**, face value **FV**, then **CPT** **PV**

Required Rate of Return vs. Yield

Required rate of return r

- The discount rate for computing the bond's value
- Chosen by investors as the opportunity cost, the return on alternatives with a similar risk
- Input r, output the value (**CPT** **PV**)

Yield y

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- Implied by the market price
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- Input r, output the value (CPT PV)
- **Subjective**

Yield y

- The discount rate for which the present value of the payments equals the price
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- Input the price, output the yield (CPT I/Y)
- **Objective**

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- Credit ratings capture default risk, investment-grade (Baa or BBB and higher) against high-yield ("junk") bonds, lower-rated bonds carry higher yields and lower prices

Module 5, Stock Valuation, Dividend Discount Model

- Stock value, the present value of the expected dividends, discounted at the required return on equity r_e

$$P = \frac{\text{Div}_1}{(1+r_e)^1} + \frac{\text{Div}_2}{(1+r_e)^2} + \frac{\text{Div}_3}{(1+r_e)^3} + \cdots$$

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- Constant dividends, $P = \text{Div}_1/r_e$. Constant growth (the Gordon growth model), $P = \text{Div}_1/(r_e - g)$
- Non-constant growth, discount the dividends year by year up to a horizon H, then the price $P_H = \text{Div}_H \times (1+g)/(r_e - g)$

Example, Valuing Non-Constant Growth

Recall from **Modules 5 and 6**

- A stock's expected dividends are in the table. Its required return on equity is 10%. Its dividends grow at 5% a year beginning in year 6. What is the value of the stock?

Year	1	2	3	4	5
Dividend	$2	$2	$2	$3	$3

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- We discounted the year-5 price of the constant-growth stream, then the five dividends, term by term,

$$\text{PV}_{\text{Price in year 5}} = \frac{3 \times 1.05}{0.10 - 0.05} \times \frac{1}{1.10^5} = \$39.12$$

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$$\text{PV}_{\text{Dividends 1–5}} = \frac{2}{1.10} + \frac{2}{1.10^2} + \frac{2}{1.10^3} + \frac{3}{1.10^4} + \frac{3}{1.10^5} = \$8.89$$

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- Total, $\text{PV}_{\text{Stock}} = 39.12 + 8.89 = \$48.00$

Example, Non-Constant Growth with a Financial Calculator



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- Enter the cash flows one by one, 0, 2, 2, 2, 3, 66
- In year 5, enter the dividend plus the price, $3+63=66$

Enter	Keys
	CF, **2nd**, **CLR Work**
0	**ENTER**, ↓
2	**ENTER**, ↓ ↓
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- Enter the cash flows one by one, 0, 2, 2, 2, 3, 66
- In year 5, enter the dividend plus the price, $3 + 63 = 66$
- **No need to discount each cash flow and sum by hand!**

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- The estimate is sensitive to the choice of multiple, and good comparables are hard to find

Example, Buying the Stock Is an NPV Decision

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- On the calculator, the same worksheet with -50 entered as the first cash flow (instead of 0)

Capital Budgeting Techniques, Summary

Criterion	Definition	Investment rule
Net present value (NPV)	Present value of benefits minus present value of costs	Take all projects with NPV > 0, reject all with NPV < 0
Internal rate of return (IRR)	The discount rate at which NPV $= 0$	Accept if IRR $>$ opportunity cost r, reject if IRR $< r$
Profitability index	Net present value divided by initial investment	Under capital rationing, rank projects by their profitability index and go down the ranking
Payback period	The number of years until the cash flows recover the initial investment	Accept if the payback period is shorter than a maximum cutoff

Facing Multiple Projects, Summary

Situation	What to do
No constraint	Take all projects with NPV $>$ 0, reject all with NPV $<$ 0
Can only choose one among mutually exclusive projects	Take the positive-NPV project with the highest NPV, not the highest IRR
Limited amount of funds for the initial investment (capital rationing)	Rank projects by their profitability index from highest to lowest and go down the ranking, stop when you run out of dollars or run out of positive-NPV projects

Thank you!

Please let me know if you have any questions.